

TRIGGER (0 to 1)	SIGNAL/POWER ON
SW	EN
EN	VOUT3, PG3V3
PG3V3	VDD2, PG2V5
PG2V5	VDD3, VOUT5, RESET, VAUD
RESET	VCNT5
VCNT5	VDD5

Bootloop protection:
Shuts off power if battery drops below 3V

Audio seems to sound best when noise on VDD2 is minimized

CPU Power sequence:
VDD2 >> VDD3 >> VDD5

GBA CPU gets very hot without 2.5V!

Sleep mode:
VCNT5 is high >> 5V is disconnected
VCNT5 is low >> 5V is connected

Audio supply turns on
after 2.5V and 3.3V
rails are stable

<https://github.com/MouseBiteLabs/Game-Boy-Enhance>

Sheet: /

File: AGBM-01_AA_1-2.kicad_sch

Title: Game Boy Enhance (AGBM-01)

Size: A4

Date: 2026-05-16

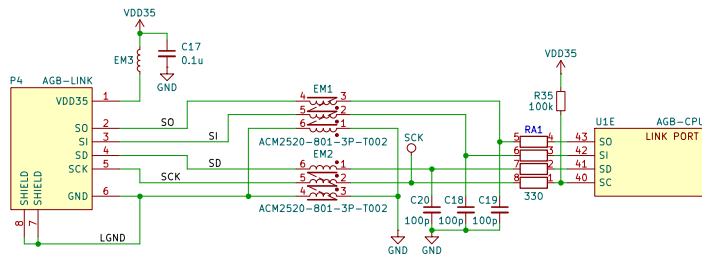
Rev: 1.2

KiCad E.D.A. 9.0.6

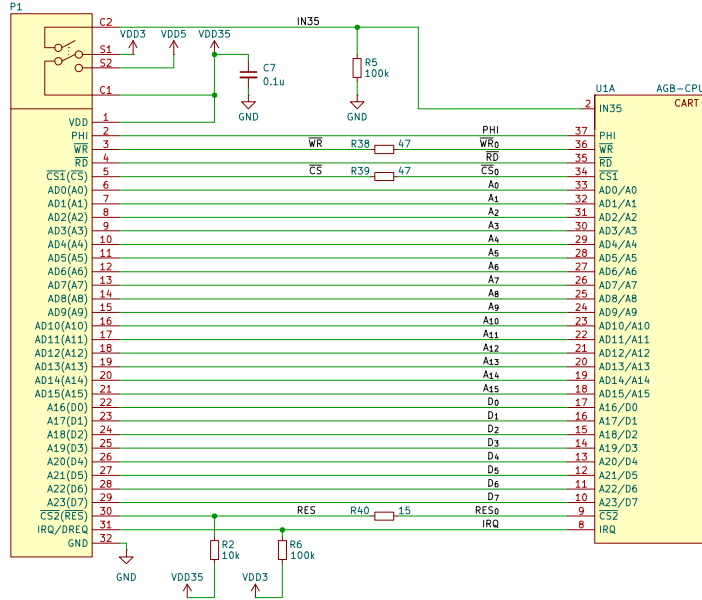
Id: 1/4

File: CPU.kicad_sch

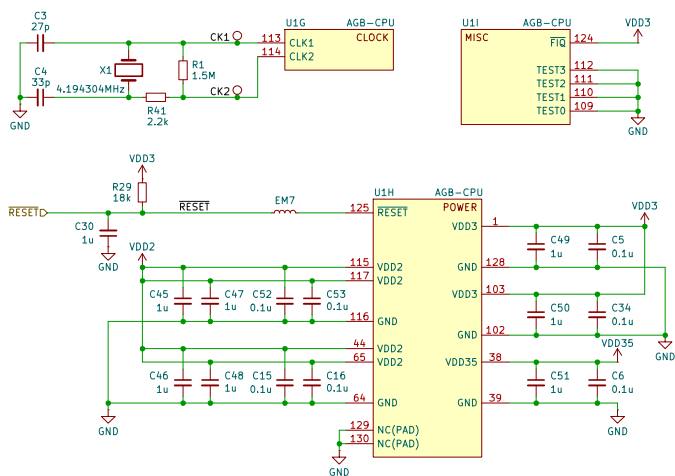
Link Port



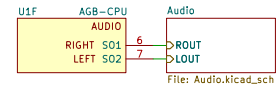
Cartridge Connector



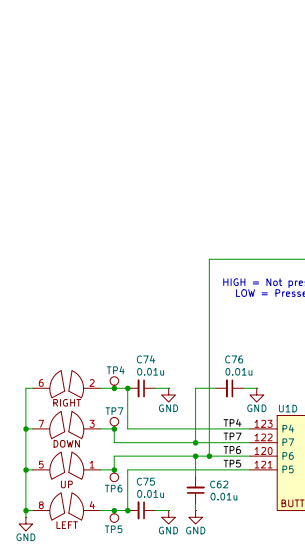
CPU Clock, Power, etc.



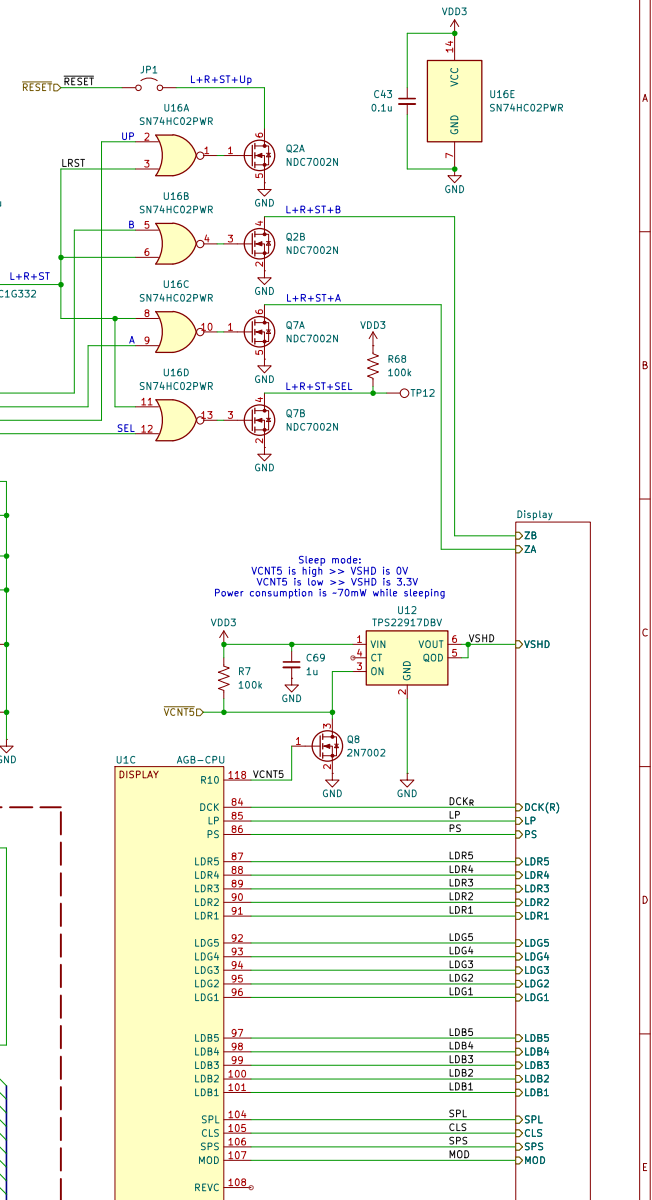
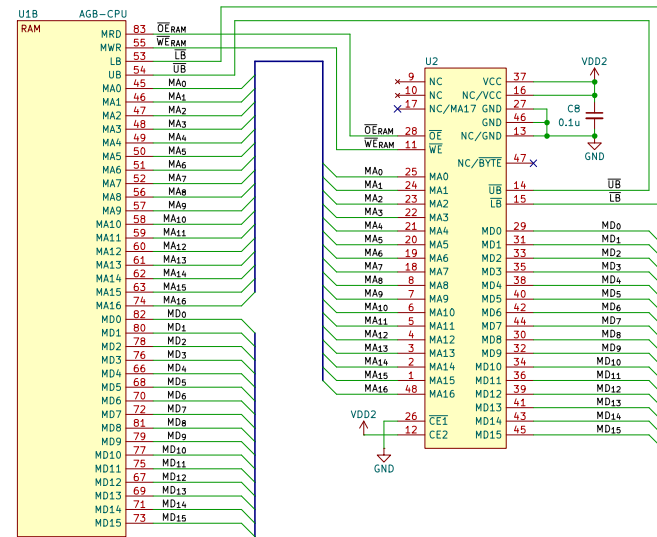
Audio



Button Inputs, IPS Screen



RAM



<https://github.com/MouseBiteLabs/Game-Boy-Enhance>

Sheet: /CPU/

File: CPU.kicad_sch

Title: Game Boy Enhance (AGBM-01)

Size: A3 Date: 2026-05-16

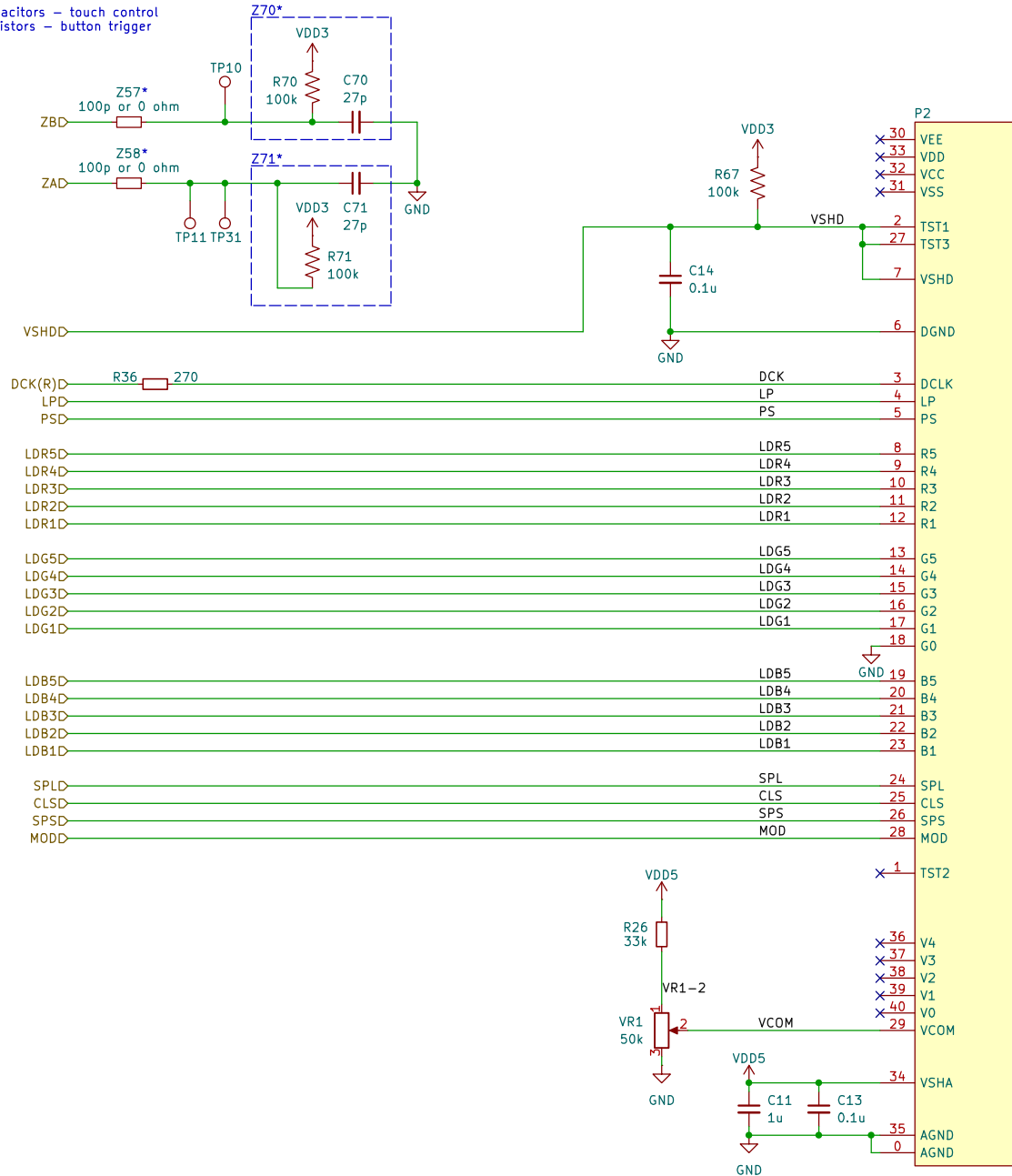
KiCad E.D.A. 9.0.6

Rev: 1.2

Id: 2/4

*Choose only ONE component!

Capacitors – touch control
Resistors – button trigger



Pins used by the laminated Hispeedido kit:

3 – DCK
6 – GND
8–17 – R, G
18 – GND
19–24 – B
26 – SPS
34 – VDD5
35 – GND

Pins used by the laminated Funnyplaying ITA kit:

3 – DCK
6 – GND
8–17 – R, G
18 – GND
19–24 – B
26 – SPS
28 – MOD
29 – VCOM
34 – VDD5
35 – GND

<https://github.com/MouseBiteLabs/Game-Boy-Enhance>

Sheet: /CPU/Display/
File: Display.kicad_sch

Title: Game Boy Enhance (AGBM-01)

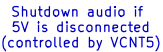
Size: A4

Date: 2026-05-16

Rev: 1.2

KiCad E.D.A. 9.0.6

Id: 3/4



Id: 4/4